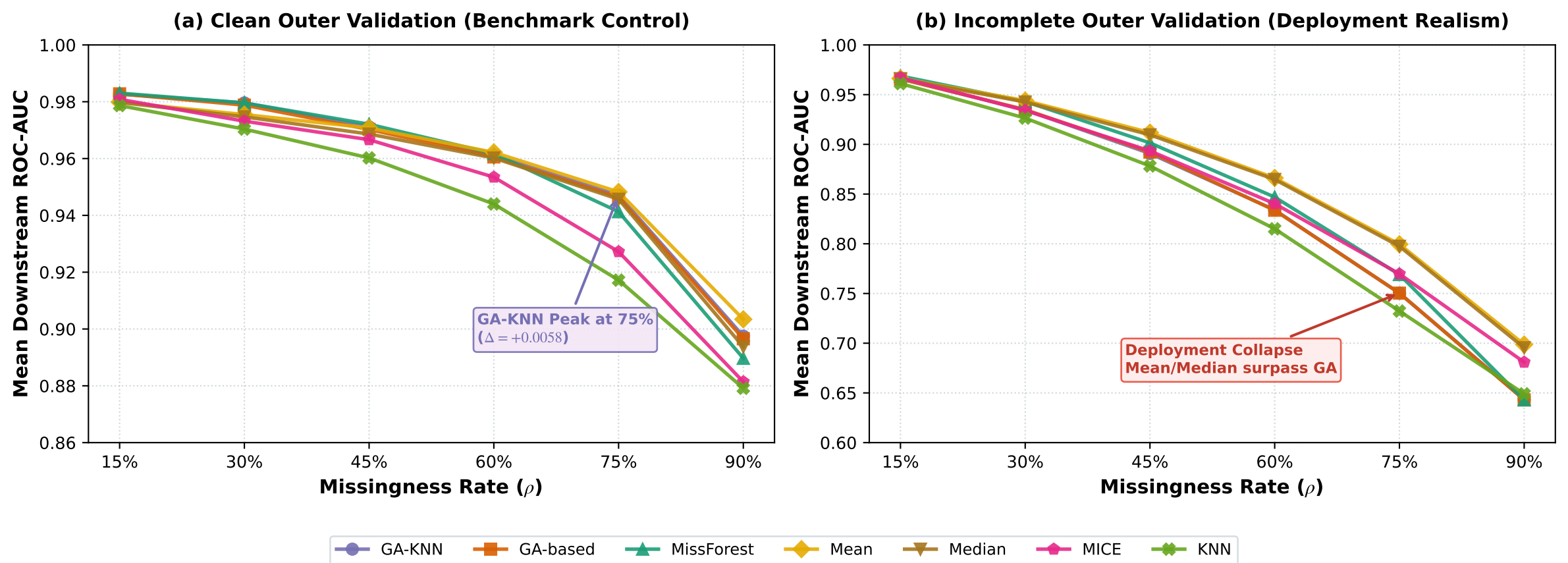
(a) Clean Outer Validation (Benchmark Control)
(b) Incomplete Outer Validation (Deployment Realism)
Mean Downstream ROC-AUC
Missingness Rate (ρ)
15%
30%
45%
60%
75%
90%
GA-KNN Peak at 75%
(Δ = +0.0058)
Deployment Collapse
Mean/Median surpass GA
GA-KNN
GA-based
MissForest
Mean
Median
MICE
KNN